

Gang Liu

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Visa Status: US F1 Visa (5-Year Valid)

Education

University of Notre Dame

Ph.D. in Electrical Engineering

Independent Adventurer

Gap Year

Beijing University of Posts and Telecommunications (BUPT)

Bachelor of Engineering in Telecommunications Engineering

University of Notre Dame

International Summer Undergraduate Research Experience (iSURE)

Jun 2026 – Expected May 2031

Supervisor: Prof. Zhi Zheng

Jun 2025 – Jun 2026

Travelling around China

Sept 2021 – Jun 2025

Avg Score: 90.75/100 (Top 1%)

Jul 2024 – Sep 2024

Supervisor: Prof. Zhi Zheng

Research Interests

Using different types of signals (RF, physiological, etc.) and various processing tools (ML, DL, etc.) to enable diverse applications (Human sensing, object sensing, etc.).

Publications

1. G. Liu, Y. Hao, Y. Zou, “CIG-MAE: Cross-Modal Information-Guided Masked Autoencoder for Self-Supervised Wi-Fi Sensing,” *IEEE Internet of Things Journal (IOTJ)*, IF: 10.62, 2026, <https://doi.org/10.1109/IOTJ.2026.3696504>.
2. G. Liu, Z. Ning, Z. Zheng, “Electrodermal Activity (EDA) for Stress Detection: A Comprehensive Survey and Benchmark,” *IEEE Transactions on Affective Computing (TAFAC)*, IF: 11.09, Submitted, [arXiv:2609.22622](https://arxiv.org/abs/2609.22622).
3. R. Yuan, S. Wang, G. Liu, and M. Peng, “Non-Line-of-Sight Ultraviolet Positioning Using Linearly-Arrayed Photon-Counting Receivers,” *IEEE Journal on Selected Areas in Communications (JSAC)*, IF: 19.92, 2024, <https://doi.org/10.1109/JSAC.2024.3413960>.
4. G. Liu, S. Wang, R. Yuan, M. Peng, “Non-line-of-Sight Optical Positioning Under Full-Duplex Ultraviolet Communication Scenarios,” *IEEE/CIC International Conference on Communications in China (ICCC)*, 2024, <https://doi.org/10.1109/ICCC62479.2024.10681787>.

Research Experience

WiFi Signal-Based Human Action Recognition (Undergraduate Thesis)

Dec 2024 – Nov 2025

Supervised by Prof. Yixuan Zou (QMUL) & Prof. Yanling Hao (QMUL) & Prof. Yuanwei Liu (Hong Kong University)

- Proposed a reconstruction-based self-supervised learning framework guided by high-value signal regions, addressing label scarcity and preserving physical semantics in WiFi sensing.
- Introduced CIG-MAE, a symmetric dual-stream masked autoencoder that reconstructs both CSI amplitude and phase and aligns cross-modal representations without negative samples.
- Outperformed state-of-the-art self-supervised methods and a fully supervised baseline across three public datasets, demonstrating strong data efficiency, robustness, and representation generalization.

Mental Stress Detection via Physiological Signal Processing

May 2024 – Present

Supervised by Prof. Zhi Zheng

University of Notre Dame

- Spearheaded the first end-to-end systematic survey on unimodal EDA stress detection; compiled a centralized resource of 26 public datasets and analyzed 22 representative studies to address field fragmentation and foster reproducibility.
- Established a standardized evaluation protocol and conducted a large-scale benchmark across 5 diverse datasets; rigorously evaluated traditional Machine Learning vs. Deep Learning models under multi-domain fusion and cross-scenario generalization settings.
- Demonstrated that feature-based Machine Learning outperforms end-to-end Deep Learning, and identified ecological validity as a critical driver of cross-domain transfer.

Wireless Ultraviolet (UV) Communication and Positioning

Aug 2023 – Jul 2024

Supervised by Prof. Renzhi Yuan

BUPT

- Designed a non-line-of-sight UV positioning system using linearly-arrayed photon-counting receivers, resolving multi-solution distance ambiguity via geometric constraints and analytical modeling.
- Developed a Fourier-series-based self-interference mitigation algorithm for full-duplex UV communication, achieving <3m positioning error and <1.2° azimuth error within 100m transceiver range.

- Conducted end-to-end modeling and MATLAB simulations integrating scattering physics, receiver geometry, and photon-counting statistics, leading to publications in *IEEE JSAC* and *IEEE ICC 2024*.

AI-Driven Spectrum Analysis via Software-Defined Radio

Aug 2023 – Jul 2024

Supervised by Prof. Yitong Liu

BUPT

- Built a wideband spectrum sensing platform (25 MHz–1.75 GHz) using RTL-SDR and MATLAB for real-time urban electromagnetic monitoring.
- Authored a spectrum analysis report for Beijing, identifying key interference patterns in Aviation/FM bands via 24-hour continuous monitoring.
- Achieved a high prediction correlation in spectrum occupancy forecasting using neural networks, enabling adaptive interference management.

Courses, Skills & Honors

- **Core Courses:** Signals and Systems, Digital Signal Processing, Principles of Communication, Electromagnetic Fields and Waves, Digital Circuit Design, Analog Circuits, Computer Networks, Machine Learning and Deep Learning, Image Recognition, Image & Video Processing, Computer Vision, Computer Graphics, Wavelet Transform
- **Languages:** Mandarin Chinese (Native), English (Proficient, **IELTS: 7.0**)
- **Programming:** C, Java, Python, MATLAB, SQL, $\text{\LaTeX}$, HTML/CSS/JS
- **Honors:** **Excellent Undergraduate Thesis (97/100, Top 1%)**; First-class Scholarship (Top 3%); Outstanding Student (Top 5%); Outstanding Student Leader (Top 5%); Second Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM)
- **Leadership & Service:** Led TF Overseas Studio (compiled BUPT's study-abroad guide and hosted alumni talks); Organized the BUPT Club Festival (1,000+ participants)